

9.2.6 Jordan Canonical Form ¶

9.2.6 Jordan Canonical Form, Part 1

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Homework 9.2.6.1. Compute the eigenvalues of $k \times k$ matrix

$$J_k(\mu) = \begin{pmatrix} \mu & 1 & 0 & \cdots & 0 & 0 \\ 0 & \mu & 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \ddots & \mu & 1 \\ 0 & 0 & 0 & \cdots & 0 & \mu \end{pmatrix} \tag{9.2.1}$$

where $k > 1$. For each eigenvalue compute a basis for the subspace of its eigenvectors (including the zero vector to make it a subspace).

[► Hint](#) [► Solution](#)

9.2.6 Jordan Canonical Form, Part 2

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The matrix in (9.2.1) is known as a Jordan block.

The point of the last exercise is to show that if A has an eigenvalue of algebraic multiplicity k , then it does not necessarily have k linearly independent eigenvectors. That, in turn, means there are matrices that do not have a full set of eigenvectors. We conclude that there are matrices that are not diagonalizable. We call such matrices defective.

Definition 9.2.6.1. Defective matrix. A matrix $A \in \mathbb{C}^{m \times m}$ that does not have m linearly independent eigenvectors is said to be defective.

Corollary 9.2.6.2. Matrix $A \in \mathbb{C}^{m \times m}$ is diagonalizable if and only if it is not defective.

Proof.

This is an immediate consequence of Theorem 9.2.5.3.

Definition 9.2.6.3. Geometric multiplicity. Let $\lambda \in \Lambda(A)$. Then the geometric multiplicity of λ is defined to be the dimension of $\mathcal{E}_\lambda(A)$ defined by

$$\mathcal{E}_\lambda(A) = \{x \in \mathbb{C}^m | Ax = \lambda x\}.$$

In other words, the geometric multiplicity of λ equals the number of linearly independent eigenvectors that are associated with λ .

Homework 9.2.6.2. Let $A \in \mathbb{C}^{m \times m}$ have the form

$$A = \left(\begin{array}{c|c} A_{00} & 0 \\ \hline 0 & A_{11} \end{array} \right)$$

where A_{00} and A_{11} are square. Show that

- If (λ, x) is an eigenpair of A_{00} then $(\lambda, \begin{pmatrix} x \\ 0 \end{pmatrix})$ is an eigenpair of A .
- If (μ, y) is an eigenpair of A_{11} then $(\mu, \begin{pmatrix} 0 \\ y \end{pmatrix})$ is an eigenpair of A .

- If $(\lambda, \begin{pmatrix} x \\ y \end{pmatrix})$ is an eigenpair of A then (λ, x) is an eigenpair of A_{00} and (λ, y) is an eigenpair of A_{11} .
- $\Lambda(A) = \Lambda(A_{00}) \cup \Lambda(A_{11})$.

▼ Solution

Let $A \in \mathbb{C}^{m \times m}$ have the form

$$A = \left(\begin{array}{c|c} A_{00} & 0 \\ \hline 0 & A_{11} \end{array} \right)$$

where A_{00} and A_{11} are square. Show that

- If (λ, x) is an eigenpair of A_{00} then $(\lambda, \begin{pmatrix} x \\ 0 \end{pmatrix})$ is an eigenpair of A .

$$\left(\begin{array}{c|c} A_{00} & 0 \\ \hline 0 & A_{11} \end{array} \right) \begin{pmatrix} x \\ 0 \end{pmatrix} = \begin{pmatrix} A_{00}x \\ 0 \end{pmatrix} = \begin{pmatrix} \lambda x \\ 0 \end{pmatrix} = \lambda \begin{pmatrix} x \\ 0 \end{pmatrix}.$$

- If (μ, y) is an eigenpair of A_{11} then $(\mu, \begin{pmatrix} 0 \\ y \end{pmatrix})$ is an eigenpair of A .

$$\left(\begin{array}{c|c} A_{00} & 0 \\ \hline 0 & A_{11} \end{array} \right) \begin{pmatrix} 0 \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ A_{11}y \end{pmatrix} = \begin{pmatrix} 0 \\ \mu y \end{pmatrix} = \mu \begin{pmatrix} 0 \\ y \end{pmatrix}.$$

- $\left(\begin{array}{c|c} A_{00} & 0 \\ \hline 0 & A_{11} \end{array} \right) \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$ implies that

$$\begin{pmatrix} A_{00}x \\ A_{11}y \end{pmatrix} = \begin{pmatrix} \lambda x \\ \lambda y \end{pmatrix},$$

and hence $A_{00}x = \lambda x$ and $A_{11}y = \lambda y$.

- $\Lambda(A) = \Lambda(A_{00}) \cup \Lambda(A_{11})$.

This follows from the first three parts of this problem.

This last homework naturally extends to

$$A = \left(\begin{array}{c|c|c|c} A_{00} & 0 & \cdots & 0 \\ \hline 0 & A_{11} & \cdots & 0 \\ \hline \vdots & \vdots & \ddots & \vdots \\ \hline 0 & 0 & \cdots & A_{kk} \end{array} \right)$$

The following is a classic result in linear algebra theory that characterizes the relationship between of a matrix and its eigenvectors:

9.2.6 Jordan Canonical Form, Part 3

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Theorem 9.2.6.4. Jordan Canonical Form Theorem. *Let the eigenvalues of $A \in \mathbb{C}^{m \times m}$ be given by $\lambda_0, \lambda_1, \dots, \lambda_{k-1}$, where an eigenvalue is listed as many times as its geometric multiplicity. There exists a nonsingular matrix X such that*

$$X^{-1}AX = \left(\begin{array}{c|c|c|c} J_{m_0}(\lambda_0) & 0 & \cdots & 0 \\ \hline 0 & J_{m_1}(\lambda_1) & \cdots & 0 \\ \hline \vdots & \vdots & \ddots & \vdots \\ \hline 0 & 0 & \cdots & J_{m_{k-1}}(\lambda_{k-1}) \end{array} \right).$$

For our discussion, the sizes of the Jordan blocks $J_{m_i}(\lambda_i)$ are not particularly important. Indeed, this decomposition, known as the Jordan Canonical Form of matrix A , is not particularly interesting in practice. It is extremely sensitive to perturbation: even the smallest random change to a matrix will make it diagonalizable. As a result, there is no practical mathematical software library or tool that computes it. For this reason, we don't give its proof and don't discuss it further.